

# The Recovery is an Illusion: Analyzing NYC Subway Ridership Patterns Post-COVID

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## Abstract

We analyzed New York City subway ridership recovery across 262 Neighborhood Tabulation Areas (NTAs) from July 2020 through December 2023, integrating MTA ridership data with PLUTO land use, Census LEHD employment, and ACS demographic datasets. Our primary contribution is the development of two distinct recovery metrics: "Bounce-back" (growth from COVID lows) versus "True Recovery" (return to pre-pandemic baselines). Our analysis reveals a critical "Recovery Illusion": while ridership has grown 2.4x from pandemic troughs, the system remains at only 72% of pre-COVID levels, with 98% of neighborhoods failing to return to baseline. Furthermore, we found that neighborhood economic factors explain 58% of the variance in Bounce-back ( $R^2 = 0.58$ ) but only 22% of True Recovery ( $R^2 = 0.22$ ). This discrepancy suggests that while the initial return to transit was driven by observable neighborhood characteristics (e.g., commercial density, education), the failure to reach full recovery is driven by unmeasured structural factors, such as employer-specific remote work policies. We present these findings via an interactive dashboard that allows stakeholders to visualize these divergent recovery trajectories.

## 1. Introduction

The COVID-19 pandemic caused an unprecedented disruption to urban transit systems, with NYC subway ridership collapsing by 85% in May 2020. As cities reopened, transit agencies and urban planners sought to predict which areas would return to normal usage levels. However, media narratives celebrating ridership growth often obscure a troubling reality: many neighborhoods remain far below their pre-pandemic baselines. This distinction is critical for equitable service allocation and long-term revenue modeling.

Current analysis is often limited by a lack of granular integration. The MTA provides aggregate statistics, and academic studies often focus on national trends or specific sectors, overlooking the neighborhood-level linkage between transit recovery and local economic indicators.

### Our contributions are as follows:

- **Dual Recovery Metrics:** We define and compare "Bounce-back" vs. "True Recovery" to expose the divergence in recovery narratives.
- **Data Integration:** We merge transit data with land use, employment, and demographic data at the NTA level.
- **Remote Work Scoring:** We implement a weighted scoring system based on NAICS industry codes to quantify "remote work potential."
- **Trajectory Clustering:** We utilize k-means clustering to identify four distinct spatial recovery patterns.
- **Predictive Divergence:** We demonstrate that "bouncing back" and "truly recovering" are fundamentally different phenomena.

## 2. Related Work

Our methodology synthesizes three streams of existing research. First, we rely on **Liu, Miller & Scheff (2020)** to establish severity baselines for pandemic transit impacts. Second, we adapt the place-based methodology of **Srinivasan et al. (2025)**, who analyzed Boston bus ridership retention; we extend this to NYC rail by incorporating land-use density variables. Third, we build on **Osorio, Liu & Ouyang (2022)**, whose findings that remote work accounted for 66% of Chicago’s ridership loss directly informed our feature engineering of the "Remote Work Potential" score using Census LEHD data.

Our work addresses a specific gap in the literature: no existing study integrates granular transit recovery data with neighborhood economic indicators while mathematically distinguishing between recovery from the trough (Bounce-back) and recovery to the baseline (True Recovery).

3. Data and Processing

We integrated four primary datasets covering the period from January 2020 to December 2023.

| Dataset               | Source        | Granularity      | Usage                         |
|-----------------------|---------------|------------------|-------------------------------|
| MTA Hourly Ridership  | data.ny.gov   | Hourly / Station | Recovery metrics & volume     |
| PLUTO Property Data   | NYC Planning  | Tax Lot          | Commercial/Residential ratios |
| LEHD Employment (WAC) | Census Bureau | Block Level      | Remote work potential scoring |
| NTA Boundaries        | NYC Open Data | NTA (262 units)  | Spatial aggregation unit      |

Data Processing Pipeline:

We processed over **270 million** hourly turnstile records, aggregating them to monthly totals and spatially joining 472 stations to 262 Neighborhood Tabulation Areas (NTAs) based on spatial containment. Using PLUTO tax lot data, we calculated commercial and residential densities, specifically isolating "office area" to target commuter behaviors. For employment data, we processed 4.2 million LEHD job records, classifying 20 NAICS industry codes into weighted "Remote Work Scores" (High/Medium/Low). The final dataset comprises 133 active subway NTAs characterized by 9 predictor features and 2 outcome variables.

4. Methodology

4.1. Recovery Metrics

We hypothesize that "recovery" is often conflated with "growth." To test this, we calculated two

metrics. The **Recovery Index (Bounce-back)** ( $\frac{Q4\ 2023\ Ridership}{Q3\ 2020\ Ridership}$ ) measures momentum from the pandemic low point, highlighting areas that emptied and refilled. The **True Recovery Index** ( $\frac{Q4\ 2023\ Ridership}{Jan-Feb\ 2020\ Ridership}$ ) measures the return to pre-pandemic baselines, serving as the critical metric for long-term sustainability.

4.2. Statistical Analysis & Clustering

We employed **Ridge Regression** (validated via 5-fold cross-validation) to model ridership changes, as it effectively penalized multicollinearity between socioeconomic variables. To categorize spatial trajectories, we applied **k-means clustering** ( $K = 4$ ), selected for interpretability and a silhouette score of 0.46. We assessed spatial dependence using **Moran's I test** (Queen contiguity).

5. Results

5.1. The "Recovery Illusion"

Our analysis reveals a stark disconnect between apparent growth and actual recovery. While the NYC subway system appears to have rebounded impressively—growing **2.37x** on average from the pandemic trough—the system remains at only **72.3%** of pre-pandemic levels. Most critically, **98% of neighborhoods (130 of 133) remain below their pre-COVID baselines**. The highest performing NTA achieved only 98% recovery, confirming that the narrative of "recovery" is largely an illusion of baseline effects.

5.2. Divergent Predictors (Regression Analysis)

The most significant finding is the contrast in model performance between the two metrics.

Table 2: Model Performance Comparison (5-Fold CV)

| Metric        | Train R2 | Test R2 | CV R2 | Interpretation     |
|---------------|----------|---------|-------|--------------------|
| Bounce-back   | 0.732    | 0.720   | 0.575 | Highly Predictable |
| True Recovery | 0.371    | 0.290   | 0.220 | Weakly Predictable |

Key Feature Drivers:

To understand *why* prediction varies so heavily, we analyzed the standardized regression coefficients.

Table 3: Top Feature Coefficients (Standardized)

| Feature           | Impact on Bounce-back ( $\beta$ ) | Impact on True Recovery ( $\beta$ ) | Insight                                                                                                                               |
|-------------------|-----------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Pct. Bachelors    | +0.38 (Very High)                 | +0.12 (Low)                         | Education drives return-to-office momentum but not full recovery.                                                                     |
| Pct. Commercial   | +0.34 (High)                      | +0.08 (Insignificant)               | Commercial zones saw mass exodus, leading to high "bounce" numbers.                                                                   |
| Median Income     | n.s                               | -0.15 (Negative)                    | Wealth doesn't predict bounce-back, but negatively impacts true recovery, likely reflecting remote work adoption in high-income jobs. |
| Remote Work Score | +0.24 (High)                      | -0.22 (Negative)                    | High remote capability actively suppresses <i>true</i> recovery.                                                                      |

**Interpretation:** Bounce-back is strongly predicted by education levels and commercial density. Wealthy, commercial, and educated neighborhoods that emptied out the most during the pandemic naturally showed the highest mathematical "bounce" when workers returned. However, **True Recovery** is nearly unpredictable ( $R^2 = 0.22$ ). The negative coefficient for Remote Work Score on True Recovery confirms that while these areas are growing, they are structurally capped by hybrid work habits.

*[Insert Figure 1 Here: Side-by-side scatter plots showing the strong correlation of features with Bounce-back vs. the weak correlation with True Recovery]*

### 5.3. Spatial Clustering and Autocorrelation

We identified four distinct recovery trajectories via k-means clustering. Moran's I testing confirmed strong spatial clustering for **Bounce-back** ( $I = 0.68$ ,  $p = 0.001$ ), with high-growth areas concentrated in Manhattan. Conversely, **True Recovery** showed weak clustering ( $I = 0.18$ ,  $p = 0.006$ ), suggesting it is driven by non-spatial factors like employer policies rather than geography.

**Table 4: Recovery Cluster Characteristics**

| Cluster       | Count | Remote Score | Avg Income | Description                                                               |
|---------------|-------|--------------|------------|---------------------------------------------------------------------------|
| 1. Near-Full  | 19    | Medium       | Moderate   | Mixed-use areas (e.g., LIC, Williamsburg) with local leisure travel.      |
| 2. Steady     | 27    | Low          | Low-Med    | Residential hubs with consistent essential transit usage.                 |
| 3. Lagging    | 54    | High         | High       | Manhattan core & wealthy suburbs; high remote work adoption.              |
| 4. Struggling | 33    | Low          | Lowest     | Outer-borough areas suffering from service cuts or economic displacement. |

*[Insert Figure 2 Here: Choropleth map of the 4 Clusters]*

## 6. Discussion

### Interpretation of the "Unpredictable" True Recovery:

The low  $R^2$  (0.22) for True Recovery indicates that land use and demographics are no longer sufficient to predict transit demand. The 28% ridership gap is likely driven by **hybrid work policies** that vary by individual employer rather than by neighborhood. Two adjacent office buildings may have vastly different occupancy mandates, creating a variance that neighborhood-level data cannot capture.

### The "Essential Paradox" & Implications:

Our data highlights a troubling equity issue. Neighborhoods with high populations of essential workers (Cluster 2 & 4) showed the *lowest* "Bounce-back" metrics because they never left the

system in 2020. Agencies prioritizing "high growth" routes risk overlooking these steady performers. Stakeholders must adjust revenue models for a structural "new normal" rather than a return to 2019 levels, and recognize that demand reduction is systemic but uneven.

## **7. Conclusion**

We analyzed NYC subway ridership recovery across 133 neighborhoods, revealing that "recovery" depends entirely on the chosen metric. While ridership successfully "bounced back" in predictable ways ( $R^2 = 0.58$ ) aligned with commercial density, the "true recovery" to pre-pandemic levels remains elusive and statistically unpredictable ( $R^2 = 0.22$ ). The remaining 28% gap is likely driven by invisible private-sector policies rather than observable neighborhood traits. Our interactive dashboard allows stakeholders to explore these divergent realities, ensuring planning is based on the reality of the new normal.

**Future Work:** Future analysis should incorporate return-to-office policy data and expand to bus networks.

## References

- [1] [MTA. "MTA Subway Hourly Ridership Data." data.ny.gov, 2024.](#)
- [2] [Liu, L., Miller, H.J., & Scheff, J. "The impacts of COVID-19 pandemic on public transit demand in the United States." \*PLOS ONE\*, 15\(11\), e0242476, 2020.](#)
- [3] [Ziedan, A., Brakewood, C., & Watkins, K. "Will transit recover? A retrospective study of nationwide ridership during COVID-19." \*Journal of Public Transportation\*, 25, 100046, 2023.](#)
- [4] [Mahfouz, M., et al. "Navigating the post-pandemic urban landscape: Disparities in transportation recovery in NYC." \*Cities\*, 2024.](#)
- [5] [Qi, Y., Liu, J., Tao, T., & Zhao, Q. "Impacts of COVID-19 on public transit ridership." \*Intl. J. Transportation Science and Technology\*, 12\(1\), 34-45, 2021.](#)
- [6] [Srinivasan, S., Shamsuddin, S., & Cheng, J. "Bus ridership retention, place-based factors, and COVID-19 in Boston." \*Transportation Research Part A\*, 196, 104479, 2025.](#)
- [7] [Che, J., Lee, J.S., & Kim, S. "How has COVID-19 impacted economic resilience of retail clusters?" \*Cities\*, 138, 104457, 2023.](#)
- [8] [Sun, F., et al. "Economic resilience during COVID-19: food retail in Seattle." \*Frontiers in Built Environment\*, 2023.](#)
- [9] [Lu, X., & Duan, Y. "Channels and countermeasures of the COVID-19 pandemic's impact on urban economic resilience." \*PLOS ONE\*, 2025.](#)
- [10] [Forouhar, A., et al. "Assessing downtown recovery rates in North American cities after COVID-19." \*Urban Studies\*, 2025.](#)
- [11] [Xiao, W., Wei, Y.D., & Wu, Y. "Neighborhood, built environment and resilience in transportation during COVID-19." \*Transportation Research Part D\*, 110, 103428, 2022.](#)
- [12] [Osorio, J., Liu, Y., & Ouyang, Y. "Executive orders or public fear: What caused transit ridership to drop in Chicago?" \*Transportation Research Part D\*, 105, 103226, 2022.](#)
- [13] [Paul, J., & Taylor, B.D. "Pandemic transit: examining transit use changes and equity implications." \*Transportation\*, 2022.](#)
- [14] [Stehle, S., & Kitchin, R. "Real-time and archival data visualisation techniques in city dashboards." \*Intl. J. Geographical Information Science\*, 34\(2\), 344-366, 2020.](#)
- [15] [Chen, B., et al. "SalienTime: User-driven Selection of Salient Time Steps for Large-Scale Geospatial Data Visualization." \*CHI 2024\*, ACM, 2024.](#)

## Appendix: Interactive Dashboard

The interactive dashboard is available at [\[future insert\]](#) and features:

- **Hero Section:** Key statistic (72% true recovery) with animated counter
- **Key Findings:** Three-card summary of main results
- **Methodology:** Four-step approach visualization
- **Ridership Timeline:** January 2020 - December 2023 with pre-COVID baseline
- **Dual Maps:** Side-by-side comparison of recovery metrics
- **Scatter Plot:** Recovery paradox visualization
- **Regression Results:**  $R^2$  comparison with coefficient details
- **Rankings:** Top and bottom neighborhoods by true recovery
- **Methodology & Limitations:** Transparency about data and methods